

Lab Experiment: 7

Title: Sending Temperature Data from NodeMCU ESP8266 to ThingSpeak Cloud

Course title: *IOT Laboratory*

Course code: *CSE-460*
4th Year 1st Semester Examination 2025

Date of Submission: 12/04/2026



Submitted to-

Dr. Md. Ezharul Islam, B.SC.(JU), M.ENG.G.(JAPAN), PHD(JAPAN)

Professor

*Department of Computer Science and Engineering
Jahangirnagar University*

&

Samsun Nahar Khandakar

Assistant Professor

*Department of Computer Science and Engineering
Jahangirnagar University
Savar, Dhaka-1342*

Class Roll	Exam Roll	Name
383	210903	Abdullah Nazmus-Sakib
379	210899	Shah Sultan
391	210911	Md Khaled Mahmud

1. Title

Sending Temperature Data from NodeMCU ESP8266 to ThingSpeak Cloud

2. Objective

- To interface **LM35 temperature sensor** with NodeMCU
- To connect NodeMCU to the internet via WiFi
- To send real-time temperature data to **ThingSpeak cloud platform**
- To visualize sensor data using an online dashboard

3. Components Required

Component	Quantity
NodeMCU ESP8266	1
LM35 Temperature Sensor	1
Micro USB Cable	1
Laptop with Arduino IDE	1
Internet Connection	1
ThingSpeak Account	1

4. Circuit Connection

◆ LM35 Sensor → NodeMCU

- Pin 1 → **3.3V**
- Pin 2 → **A0 (Analog Pin)**
- Pin 3 → **GND**

 As described in your lab sheet (page 1), LM35 is connected directly to A0 for analog reading

5. Program (Arduino Code)

```
#include <ESP8266WiFi.h>
#include <WiFiClient.h>
#include <ThingSpeak.h>

const char* ssid = "YOUR_WIFI";
const char* password = "YOUR_PASSWORD";

int val;
int pin = A0;

WiFiClient client;
```

```
unsigned long myChannelNumber = YOUR_CHANNEL_ID;
const char * myWriteAPIKey = "YOUR_API_KEY";

void setup() {
  Serial.begin(9600);
  delay(10);

  WiFi.begin(ssid, password);
  ThingSpeak.begin(client);
}

void loop() {
  val = analogRead(pin) * 0.322265;

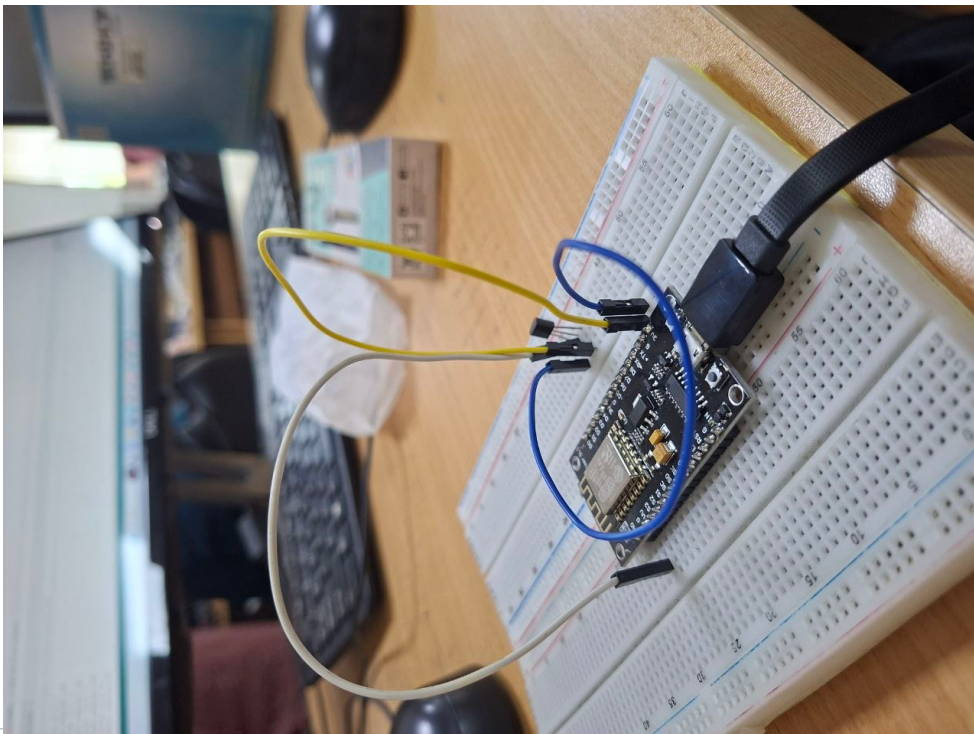
  Serial.print("Temperature: ");
  Serial.print(val);
  Serial.println("*C");

  ThingSpeak.writeField(myChannelNumber, 1, val, myWriteAPIKey);

  delay(15000); // Required delay for ThingSpeak
}
```

6. Working Principle

- LM35 sensor outputs analog voltage proportional to temperature
- NodeMCU reads this analog value via A0 pin
- Value is converted into temperature (°C)
- NodeMCU connects to WiFi network
- Data is sent to ThingSpeak server using API key
- ThingSpeak stores and displays data as a graph



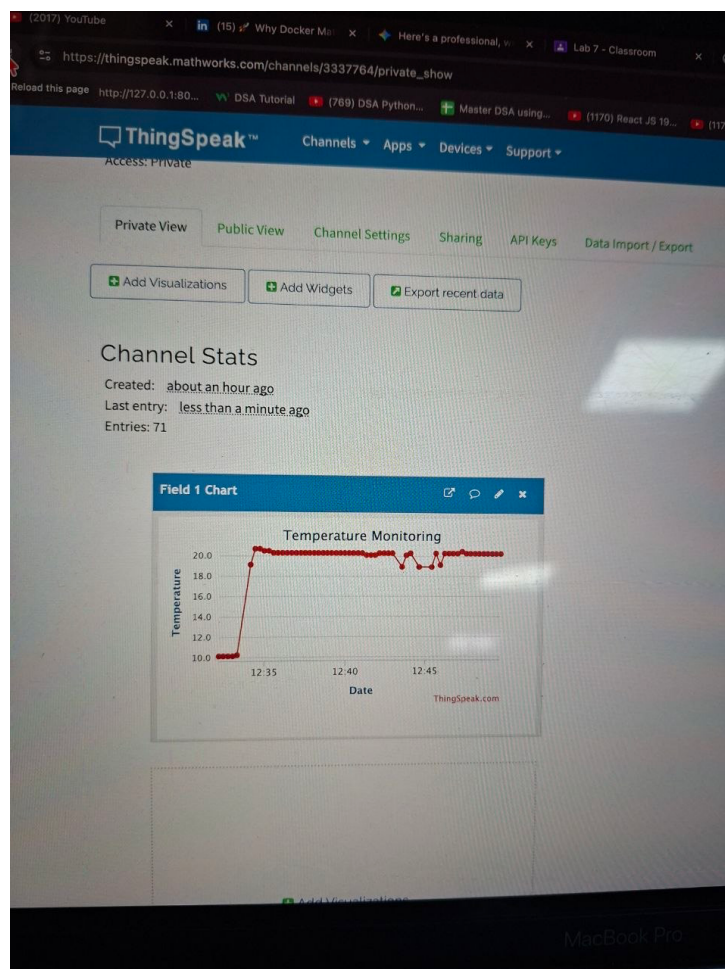
7. Procedure

1. Collect all components
2. Connect NodeMCU to laptop via USB
3. Install ESP8266 board package in Arduino IDE
4. Select **NodeMCU 1.0 (ESP-12E Module)**
5. Connect LM35 sensor properly
6. Create ThingSpeak account and channel
7. Get Channel ID and Write API Key
8. Install ThingSpeak library
9. Write and upload the program
10. Open Serial Monitor to check readings
11. Open ThingSpeak dashboard to view graph

8. Observation

- Serial Monitor displayed temperature values
- Data was successfully uploaded to ThingSpeak
- Graph updated automatically in real-time
- Temperature values changed according to environment

9. Result



The temperature data from LM35 sensor was successfully transmitted to the ThingSpeak cloud platform using NodeMCU ESP8266 and displayed in graphical form in real-time.

10. Applications

- Smart weather monitoring systems
 - Remote temperature monitoring
 - Industrial IoT systems
 - Smart agriculture
 - Home automation
-

11. Precautions

- Ensure correct WiFi credentials
 - Use correct API key and Channel ID
 - Maintain stable internet connection
 - Add proper delay (≥ 15 seconds)
 - Check wiring carefully
-

12. Conclusion

This experiment demonstrated the integration of sensors, microcontrollers, and cloud platforms in IoT systems. It provided practical knowledge of real-time data acquisition, wireless communication, and cloud-based monitoring, which are essential for modern IoT applications.